



The internet is the primary source of all the information and communication in today's modern world. A website is the face of the internet. Let's learn how websites are made.



Communication is a vital part of our daily life. We express our thoughts, ask questions or explain things using various languages like English, Hindi, etc. Websites communicate with the internet using a specific language called HTML.

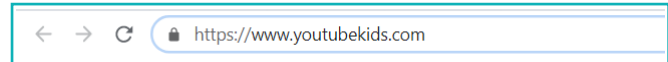
The languages we speak have certain rules for forming sentences, questions, poems, etc. We follow these rules so that our word arrangements make sense to other people. Similarly, the web language HTML has to follow certain rules too. These rules are called protocol. Rules designed for communication are called Hyper Text Transfer Protocol (HTTP).



HTTP (Hyper Text Transfer Protocol): A communications protocol used by the World Wide Web to transfer webpages and files over the internet.

HTTPS: A website contains important data like username, passwords or information that must be protected and secured. Therefore, a secure extension of HTTP called HTTPS (Hyper Text Transfer Protocol **Secure**) is used by most websites nowadays.

For example: When a website is opened in the browser, the http or https is a part of the domain name of every website.



HTML

A special language is used to create webpages and communicate over the web to display website data.

HTML (Hyper Text Markup Language) is the communication language used between web browsers and servers to organise the contents of a website in a structure. It helps browsers understand and display website files. All websites are structured using HTML.



Just like a building structure, the HTML language builds a website's structure. It contains elements that tell a browser about the structure and type of content on the webpage.



Exercise

Tick mark ☒ the correct answer

1 The secure communication protocol used for the internet is called:

- a. HTTP ☐ b. HTTPS ☐ c. HTML ☐ d. DNS ☐

2 HTML creates the _____ of a website.

- a. Structure ☐ b. Design ☐ c. Backend ☐ d. None of the above ☐

We will learn more about the HTML language and use it to create a webpage.



Activity

■ Create a webpage of your personal information, your hobbies, idols, your art and projects.

A personal information webpage is called a portfolio. It is created to showcase skills and achievements. It helps in college admissions.

To create a webpage like this, we require a code editor.

Note:

Code Editor: Code editor is a software where we write and run the code and view the output.

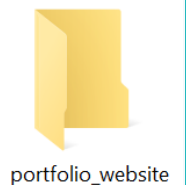
Activity

We will use Visual Studio code editor for web development.

1 Install software

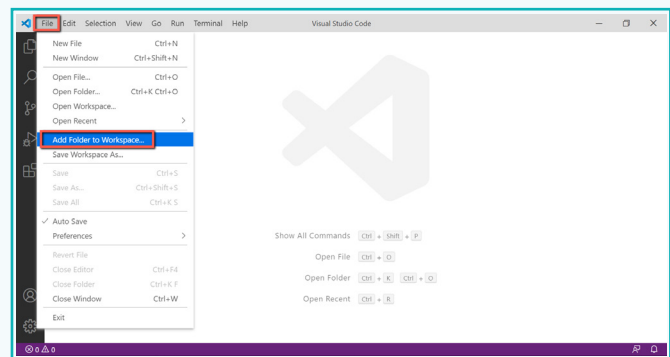
- After installing the software, create a folder with an appropriate project name to save project files.

It is a good practice to create a separate project folder and save all related files in the same location.



2 Open VS editor and create new file

- Open the VS editor.
- Click on File and select Add project folder from the dropdown.

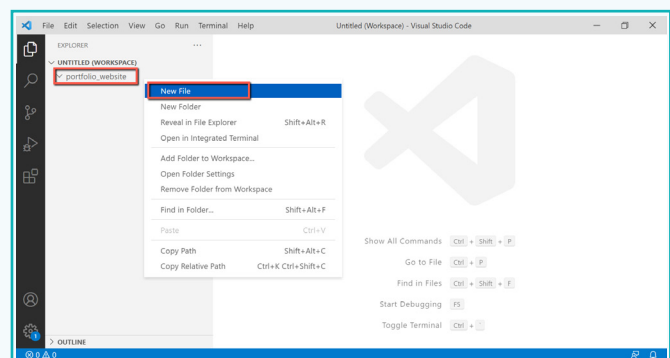


- Select the project folder that we just created.

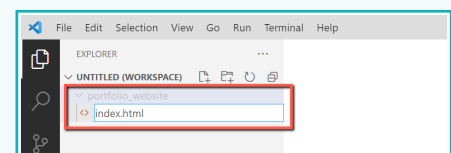
A few other windows may be open, such as welcome, welcome guide etc. Close them, we don't need them right now.

The project folder is on the left side of the screen.

- To add a new editor file, right click on project folder and click on new file.

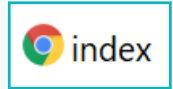


To create an HTML document, the file must have a '.html' extension, just like an audio file has '.mp3' or an image file has '.png'. The main page of a website is always named the index. It is a good practice to use lowercase letters to name webpages.



Activity

If a project folder is opened, a new file “index.html” will be visible with the icon of the default web browser.



If we double click on the file, it will open the webpage in the browser.

3 Create HTML5 file

a. Open the editor and type html in the first line, select html5 option. This will provide the latest template as per HTML5.

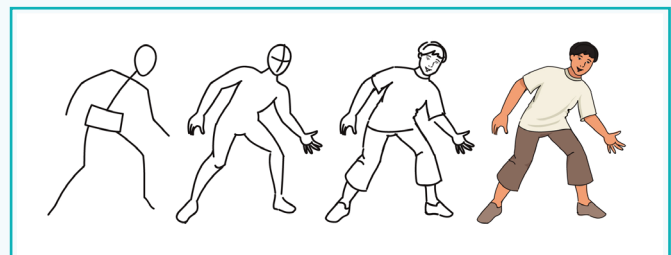


b. Press Enter and a code template will appear. This is the basic HTML code structure that every webpage has. Let us understand this in more detail.



HTML Code Structure

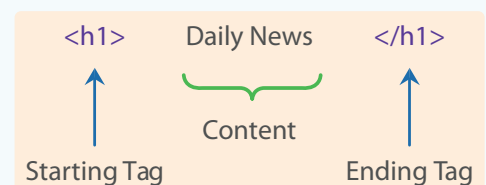
When we draw a human figure, a basic stick drawing is drawn as its base structure. This way, the image has the correct shape and form.



Similarly, to create a webpage, a standard HTML code is used as a basic structure. This code structure has many HTML elements with specific functions.

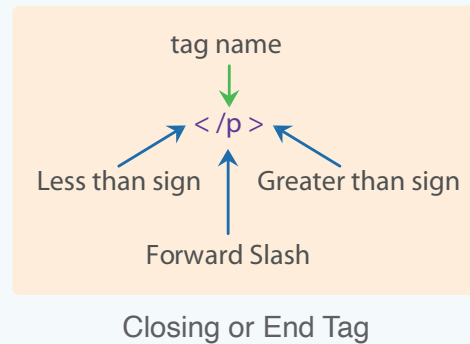
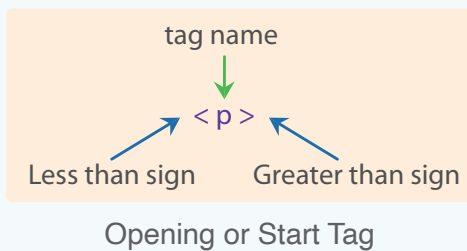
HTML elements: An HTML element is a component of code used to display a type of content of an HTML document in browser. It is comprised of a start tag, an end tag and content in between.

For example: Heading element h1 is used to display headings in the document.



Activity

HTML Tags: HTML tags are a set of characters that make up the start and end part of an HTML element. HTML tags have a specific format. Their structure is as follows:



Some types of HTML tags are:

- Head <head> </head>
- Body <body> </body>
- Title <title> </title>
- Heading <h1> </h1>, <h2></h2>....<h6></h6>
- Paragraph <p> </p>

All of them tell the browser about the type of content inside the tag. Some tags only have an opening tag and no end tag. They are called self-closing tags. Some examples of these are <hr>,
, etc.

HTML Attributes: Attributes are part of an element and are used to customize the properties of a tag. Attributes are defined within the opening tag.

For example: <hr> is a single starting tag that draws a horizontal line. The line's thickness and colour can be changed with attributes.

```
<hr size="3" color="black">
```

The format of attributes is important. Attribute values are always mentioned between double inverted commas " ".

Attribute_name = " attribute_value"
size="3" or color="black"

Let us look at the basic code structure of an HTML code.

```

1  <!DOCTYPE html>
2
3  <html>
4
5    <head>
6      <meta charset="UTF-8">
7      <meta name="viewport" content="width=device-width,initial-scale=1.0">
8      <title> </title>
9    </head>
10
11   <body>
12
13   </body>
14
15 </html>
  
```

Activity

<!DOCTYPE html> Document Type Declaration: This is a single tag without an end tag. It tells the browser that the document version is HTML 5.

`<html> ... </html>` always envelopes all code to indicate that the code inside these tags is HTML.

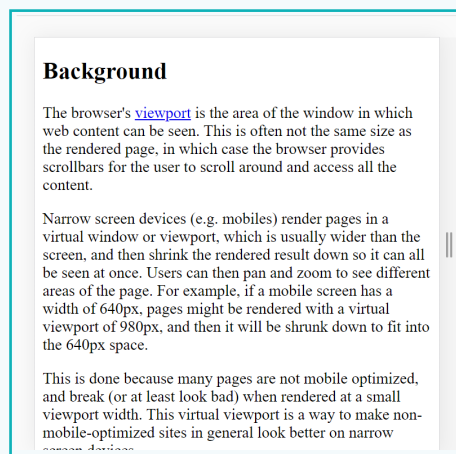
`<head> ... </head>` envelopes code which sends important information to the browser so that different browsers can display the same page uniformly. Code inside head elements does not include any content to be displayed on the website. Some code inside head elements is also used by search engines to find websites with the help of keywords.

In a basic HTML structure, two important tags included inside head tags are meta and title

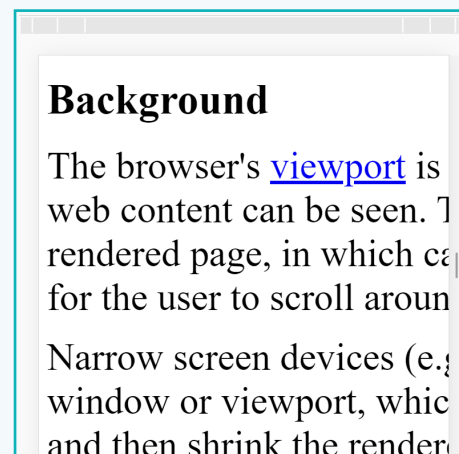
Meta tag: Meta is self-closing tag that tells the browser important information or metadata about the webpage. This includes the character set, content type, author, description, and keywords. There are two important meta tags.

`<meta charset="utf-8">` tag adds a set of characters like “a,b,...,A,B,...,@,#, ...” etc. to display on a webpage.

`<meta name="viewport" content="width=device-width, initial-scale=1.0">` tag controls how a webpage looks in smaller devices like smartphones. Here is an example of a webpage on a phone, with and without a viewport meta tag.



With viewport meta tag

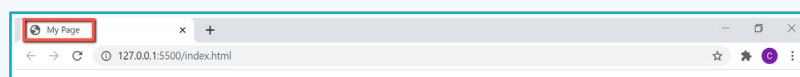


Without viewport meta tag

Title element: We write the page title using Title element.

`<title>My Page</title>`

This title is displayed on the window or tab at the top of the browser.



Activity

4 Add title of the webpage

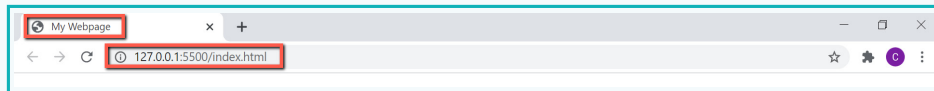
- Add the website title 'my webpage' in the head section.

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
5 <meta http-equiv="X-UA-Compatible" content="IE=edge">
6 <meta name="viewport" content="width=device-width, initial-scale=1.0">
7 <title> My Webpage </title>
8 </head>
9 <body>
10
11 </body>
12 </html>

```

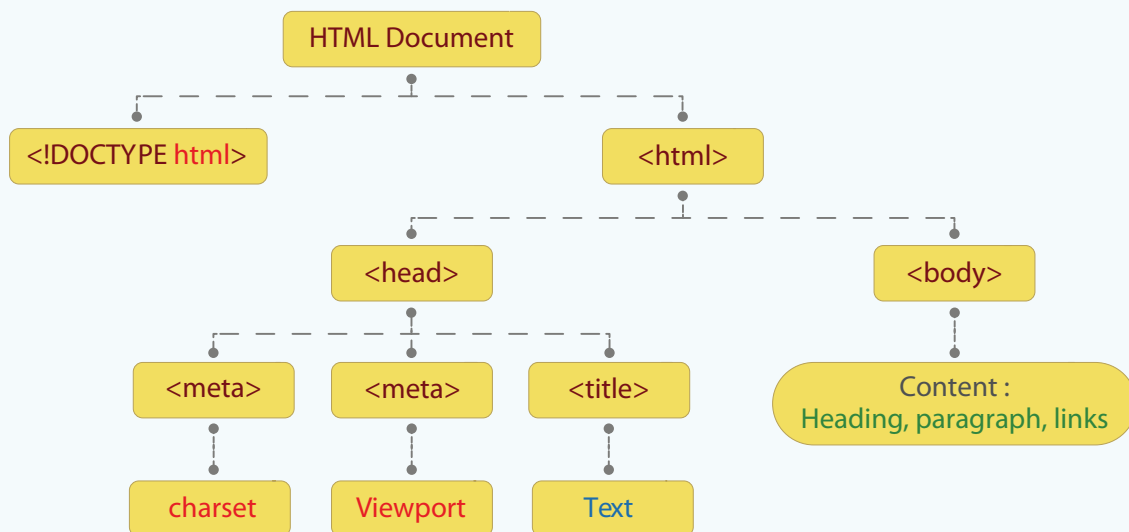
We can create a local server and check the output in the browser. To do this, press **Alt+ I + o**. The page will open in the computer's default browser. Note that, Live Server extension should be installed on VS Code.



The text 'My Webpage' in the title element appears as the name of the tab. The file name 'index.html' will appear at the end of the website address. We will add content in body tags for the webpage.

`<body> ... </body>` is a tag that contains code used to display webpage content.

We can visualize this structure as a tree diagram:



Here, 'HTML Document' is the root and has two branches: `<!DOCTYPE html>` and `<html>`. `<html>` has two more branches: `<head>` and `<body>`. `<head>` is further divided into branches according to tags used for the search engine. `<body>` has tags which are required to display content on the web page.

Exercise

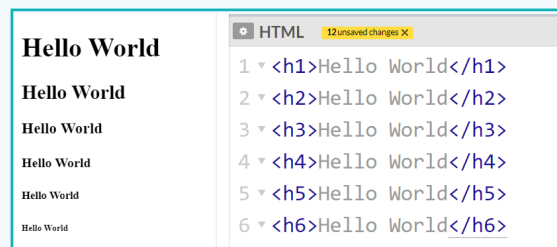
Fill in the blanks

- 3 Every HTML code starts with _____.
- 4 _____ element contains the code that tells a browser important information about the page and its structure.
- 5 _____ tag is used to add the character set used in webpage.

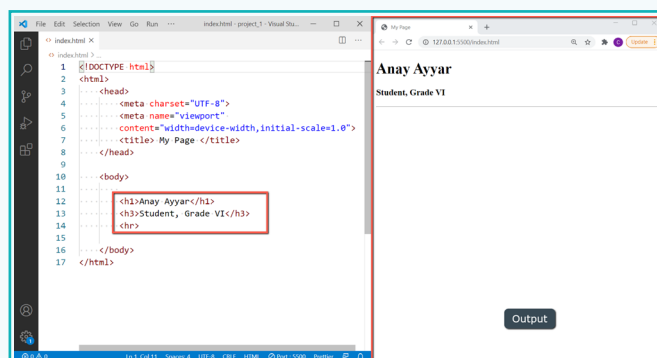
Activity

In our activity, we will create our own portfolio, hence the first element should be our name. For this, we will use the tag `<h1>`.

Heading `<h1>` to `<h6>` are heading tags that display text inside them as headings. There are 6 heading elements from 'h1' to 'h6' with descending text sizes.



- 5 Add a heading and sub-heading for the webpage
 - a. Add h1 element with your name as the heading.
 - b. Add a small heading below your name stating the grade you study in or who you aspire to be. Use heading tags h3 or h4.
 - c. Use the tag `<hr>` to add a horizontal line after the heading to end the section.

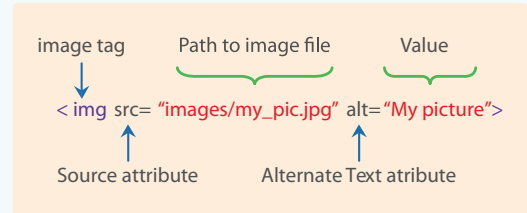


- 6 Insert an image in the webpage

Images are important components in websites. They make a website visually appealing and engaging. We can use an image tag `<img>` to add images to a webpage.

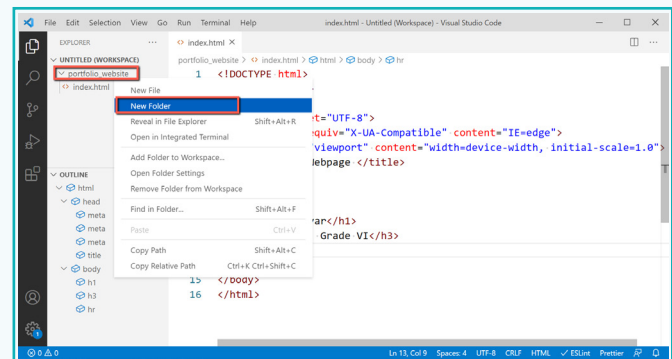
Activity

Image tag: `<img>` is a self-closing tag used to display images on a webpage. Source attribute 'src' is used to add the image path. 'alt' attribute is used to add alternate text if the image cannot be displayed.

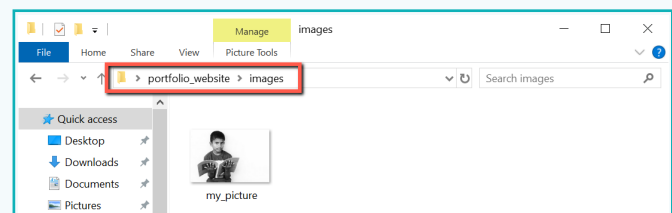


It is a standard practice to store all images for the website in a separate folder inside the project folder. We can call it 'images'.

- a. Right click on the project folder and click on New Folder.

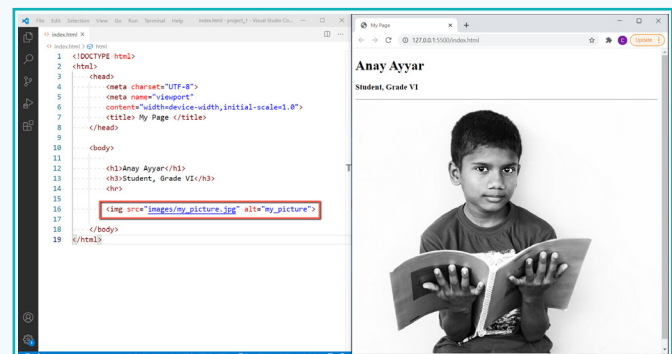


- b. Name the folder 'images' and add all images required for website.



- c. Add tag `<img src="" alt="">`.
 d. Add the image path in source attribute as "images/my_picture.jpg". Without providing the correct location of the image, we cannot use it.

- e. Add the 'alt' attribute value as 'my picture' to add a description of this picture. If an image fails to be visible because of any reason, the text inside "alt" appears here.



Activity

Note:

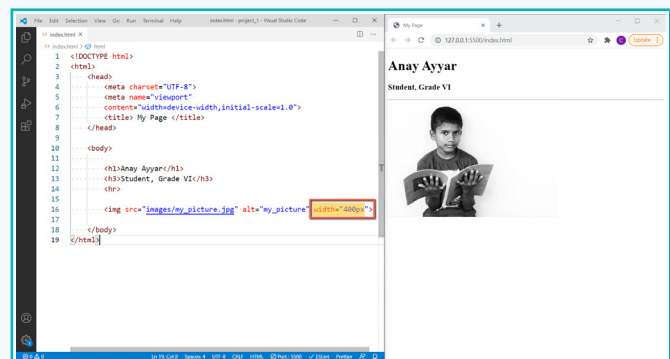
If the image folder is stored in the same location as the index.html file, the image can be added with

src = "folder_name/image_name.extention"

For example: in this case, src = "images/my_picture.jpg"

We can see that the image in the output looks too big compared to the page. We can use height or width attribute to specify the image size.

- f. Add attribute width = "400px" in tag to set width to 400 pixels.



Exercise

Answer in one line

- 6 What happens if width attribute is written in the next line? Try it out and observe.

```

```



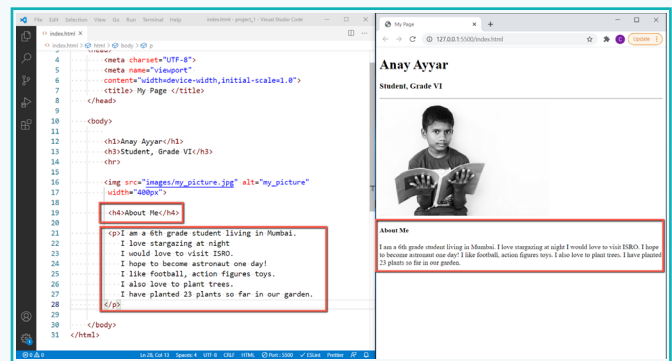
Activity

7 Write a paragraph for the webpage

We have added a name, brief information and an image. Now, we will add an introduction about ourselves with a section called 'about me'.

Paragraph <p> </p> tag: This element is used to add a text paragraph. It has a start as well as an end tag.

- Add a small sized heading 'About me'.
- Add a short text about yourself using a paragraph tag.



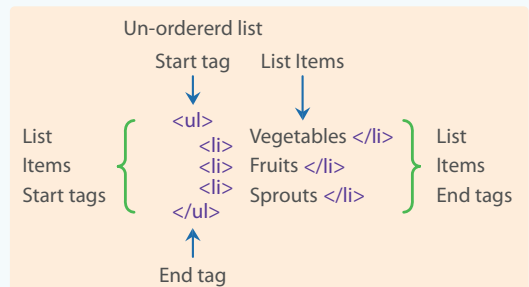
It can be observed that even if sentences are written in separate lines in <p> tag, they appear continuous on the webpage. Any content in <p> tag is considered one paragraph. To add content in the next line,
 tag is used.

8 Create list

Now, we want to list down our hobbies. For that, we will use and tags.

Unordered List and List Items: tags are used to create lists with bullet points. Each item in the list is created using 'list item' tags which are enveloped by tags.

For example:

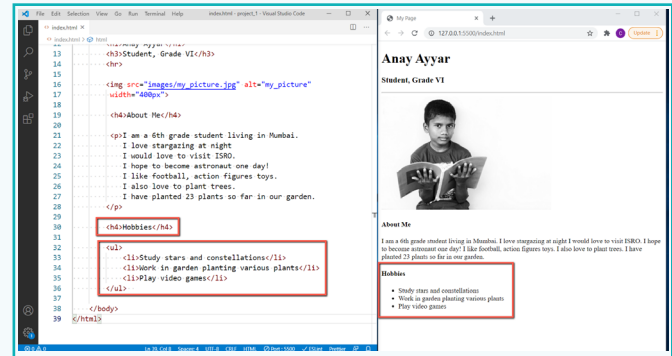


To add hobbies, we will use the list method.

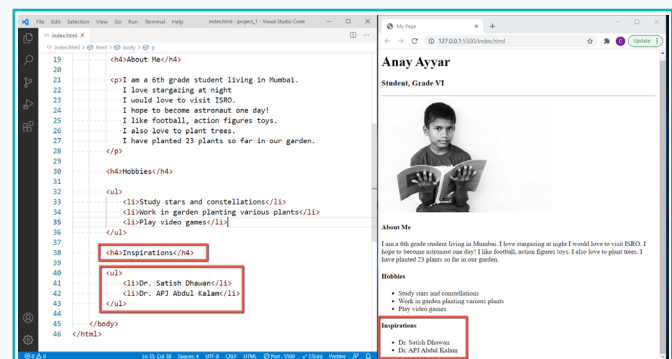
- Add a small sized heading "Hobbies".
- Add tags.

Activity

- c. Add `<li>` `</li>` tags inside `<ul>` tags to describe each hobby on a separate line.



Who are your inspirations? Create a small sized heading “Inspirations” and use `<ul>` and `<li>` elements to make a list of the people who inspire you.



In this lesson, we have created our first ever web page. We learnt about HTML code structure, `<head>` tags and other tags that are used under the `<head>` tag. We also saw the `<body>` tag and customized information using various tags like heading, list, and image.

You can continue to add more points to your portfolio using all the HTML elements we have learned so far.

Exercise

Tick mark ☒ the correct answer

7 _____ element is used to display a title or a heading.

a. Head ☐

b. H1-h ☐

c. P ☐

d. hr ☐

8 tag is used along with _____ tag for creating lists.

a. <p> </p> ☐

b. <h1> </h1> ☐

c. ☐

d. <hr> ☐

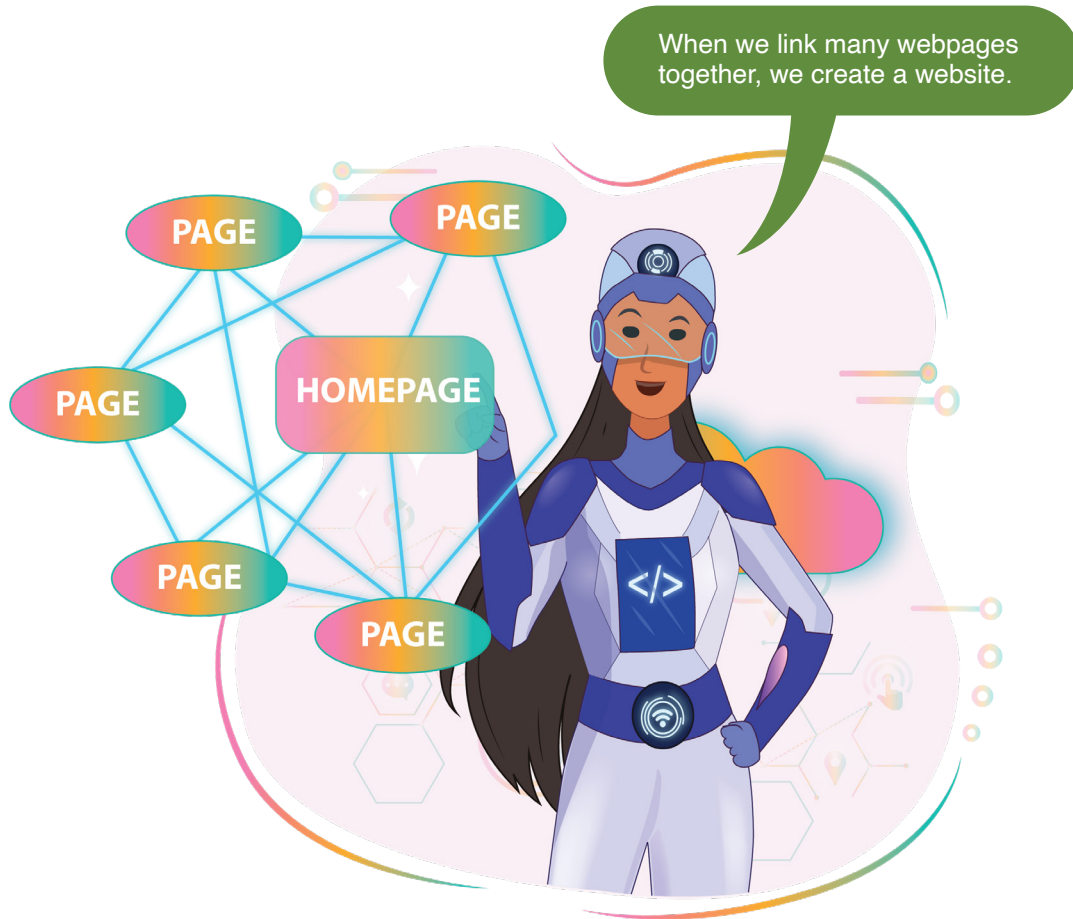
9 _____ attribute in element stands for “image source”.

a. Alt ☐

b. Src ☐

c. Width ☐

d. None of them ☐



Tech Flash

- **HTTP** : a communications protocol used to transfer the data over the internet.
- **HTTPS** : secure version of HTTP used to protect data over the internet.
- **HTML** : a language used to create websites and help browsers display content over the web.
- **HTML Element** : a component of code used to display a specific type of content in an HTML document.
- **HTML Tags** : a set of characters that format a part of the HTML element in a specific manner.
- **HTML Attributes** : part of an element's opening tag and are used to customise its properties.